

4.16 DESIGN RULES FOR FLANGED JOINTS

4.16.1 SCOPE

4.16.1.1 The rules in 4.16 shall be used to design circular flanges subject to internal and/or external pressure. These rules provide for hydrostatic end loads, gasket seating, and externally applied axial force and net-section bending moment.

4.16.1.2 The rules in 4.16 apply to the design of bolted flange connections with gaskets that are entirely located within the circle enclosed by the bolt holes. The rules do not cover the case where the gasket extends beyond the bolt hole circle or where metal-metal contact is made outside of the bolt circle.

4.16.1.3 It is recommended that bolted flange connections conforming to the standards listed in 4.1.11 be used for connections to external piping. These standards may be used for other bolted flange connections and dished covers within the limits of size in the standard and pressure-temperature ratings permitted in 4.1.11. The ratings in these standards are based on the hub dimensions given or on the minimum specified thickness of flanged fittings of integral construction. Flanges fabricated from rings may be used in place of the hub flanges in these standards, provided that their strength and rigidity, calculated by the rules in this paragraph, are not less than that calculated for the corresponding size of hub flange.

4.16.1.4 The rules of this paragraph should not be construed to prohibit the use of other types of flanged connections, provided they are designed in accordance with Part 5.

4.16.2 DESIGN CONSIDERATIONS

4.16.2.1 The design of a flange involves the selection of the flange type, gasket material, flange facing, bolting, hub proportions, flange width, and flange thickness. The flange dimensions shall be selected such that the stresses in the flange and the flange rigidity satisfy the acceptability criteria of this paragraph.

4.16.2.2 In the design of a bolted flange connection, calculations shall be made for the following two design conditions, and the most severe condition shall govern the design of the flanged joint.

(a) *Operating Conditions.* The conditions required to resist the hydrostatic end force of the design pressure and any applied external forces and moments tending to part the joint at the design temperature.

(b) *Gasket Seating Condition.* The conditions existing when the gasket or joint-contact surface is seated by applying an initial load with the bolts during assembly of the joint, at atmospheric temperature and pressure.

4.16.2.3 Calculations shall be performed using dimensions of the flange in the corroded and uncorroded conditions.

4.16.2.4 In the design of flange pairs, each flange is designed for its particular design loads of pressure and gasket reactions. The bolt load used to design each flange, however, is that load common to the flange pair and equal to the larger of the bolt loads calculated for each flange individually. No additional rules are required for design of flange pairs. After the loads for the most severe condition are determined, calculations shall be made for each flange following the rules of this paragraph.

4.16.2.5 In the design of flange pairs where pass partitions with gaskets are used, the gasket loads from the partition(s) shall be included in the calculation of bolt loads. Partition gaskets may have different gasket constants than the ring gasket inside the bolt circle. In the design of flanges with noncircular gaskets or with partitions of any shape, gasket reactions from all surfaces with gaskets shall be included in calculating bolt loads.

4.16.3 FLANGE TYPES

(25)

4.16.3.1 For the purpose of computation, there are two major categories of flanges:

(a) *Integral Type Flanges.* This type covers designs where the flange is cast or forged integrally with the nozzle neck, vessel or pipe wall, butt welded thereto, or attached by other forms of welding such that the flange and nozzle neck, vessel or pipe wall are structurally equivalent to integral construction. Integral flanges shall be designed considering structural interaction between the flange and the nozzle neck, vessel, or pipe wall, which the rules account for by considering the neck or wall to act as a hub. Integral type flanges are referenced below. The design flange and bolt loads are shown in Figure 4.16.1.

(1) Integral type flange without a hub - Figure 4.16.1, sketches (a) and (b) and Table 4.2.9, Details 2, 4, 9, 10, and 11

(2) Integral type flanges where $g_1 = g_o$ - Figure 4.16.1, sketch (c)

(3) Integral type flanges with a hub - Figure 4.16.1, sketches (d), (e), and (f) and Table 4.2.9, Details 6, 7, and 8

(4) Integral type flanges with nut stops - Figure 4.16.3 and Figure 4.16.4

(b) *Welded Slip-on Type Flanges.* This type covers designs where the flange consists of a cast or forged ring with a hub, which is installed over the nozzle neck, vessel, or pipe wall and attached by welding at both the hub and face ends of the flange. The design flange and bolt loads are shown in Figure 4.16.2 and Table 4.2.9, Details 1, 3, and 12.

(c) *Loose Type Flanges.* This type covers those designs in which the flange has no integral connection to the nozzle neck, vessel, or pipe wall. The design flange and bolt loads are shown in Figure 4.16.5 and Table 4.2.9, Detail 5.

4.16.3.2 The integral and loose type flanges described above can also be applied to reverse flange configurations. Integral and loose type reverse flanges are shown in Figures 4.16.6 and 4.16.7.

4.16.4 FLANGE MATERIALS

4.16.4.1 Materials used in the construction of bolted flange connections, excluding gasket materials, shall comply with the requirements given in Part 3.

4.16.4.2 Flanges made from ferritic steel shall be given a normalizing or full-annealing heat treatment when the thickness of the flange, t (see Figures 4.16.1 through 4.16.6), exceeds 75 mm (3 in.).

(25) **4.16.4.3** Fabricated flanges with hubs shall be in accordance with the following:

(a) Flanges with hubs shall not be machined from plate or bar (except as permitted in 3.2.5.2) material unless the material has been formed into a ring and the following additional conditions are met:

(1) In a ring formed from plate, the original plate surfaces are parallel to the axis of the finished flange.

(2) The joints in the ring are welded butt joints that conform to the requirements of Part 6. The thickness to be used to determine postweld heat treatment and radiographic requirements shall be $\min[t, (A - B)/2]$.

(3) The back of the flange and outer surface of the hub shall be examined by either the magnetic particle method or the liquid penetrant method in accordance with Part 7.

4.16.4.4 Bolts, studs, nuts, and washers shall comply with the requirements of Part 3 and referenced standards. It is recommended that bolts and studs have a nominal diameter of not less than 12 mm (0.5 in.). If bolts or studs smaller than 12 mm (0.5 in.) are used, then ferrous bolting material shall be of alloy steel. Precautions shall be taken to avoid overstressing small-diameter bolts. When washers are used, they shall be through hardened to minimize the potential for galling.

4.16.5 GASKET MATERIALS

4.16.5.1 The gasket constants for the design of the bolt load (m and y), are provided in Table 4.16.1. Other values for the gasket constants may be used if based on actual testing or data in the literature, as agreed upon between designer and the user.

4.16.5.2 The minimum width of sheet and composite gaskets, N , is recommended to be no less than that given in Table 4.16.2.

NOTE: Gasket materials should be selected that are suitable for the design conditions. Corrosion, chemical attack, creep and thermal degradation of gasket materials over time should be considered.

4.16.6 DESIGN BOLT LOADS

(25) **4.16.6.1** The procedure to determine the bolt loads for the operating and gasket seating conditions is shown below.

Step 1. Determine the design pressure and temperature of the flange joint.

Step 2. Select a gasket and determine the gasket factors m and y from Table 4.16.1, or other sources. The selected gasket width should comply with the guidelines detailed in Table 4.16.2.

Step 3. Determine the width of the gasket, N , basic gasket seating width, b_0 , the effective gasket seating width, b , and the location of the gasket reaction, G , based on the flange and gasket geometry, the information in Table 4.16.3 and Figure 4.16.8, and the equations shown below. Note that for lap joint flanges, G is equal to the midpoint of contact between the flange and the lap, see Figures 4.16.5 and 4.16.8.

(a) For $b_0 \leq 6$ mm (0.25 in.), G is the mean diameter of the gasket contact face and

$$b = b_0 \quad (4.16.1)$$

(b) For $b_0 > 6$ mm (0.25 in.)

$$b = 0.5C_{ul}\sqrt{\frac{b_0}{C_{ul}}} \quad (4.16.2)$$

$$G = G_c - 2b \quad (4.16.3)$$

Step 4. Determine the design bolt load for the operating condition.

$$W_o = 0.785G^2P + 2b\pi GmP \text{ for non-self-energized gaskets} \quad (4.16.4)$$

$$W_o = 0.785G^2P \text{ for self-energized gaskets} \quad (4.16.5)$$

Step 5. Determine the design bolt load for the gasket seating condition.

$$W_g = \left(\frac{A_m + A_b}{2} \right) S_{bg} \quad (4.16.6)$$

The parameter A_b is the actual total cross-sectional area of the bolts that is selected such that $A_b \geq A_m$, where:

$$A_m = \max \left[\left(\frac{W_o + F_A + \frac{4M_E}{G}}{S_{bo}} \right), \left(\frac{W_{gs}}{S_{bg}} \right) \right] \quad (4.16.7)$$

$$W_{gs} = \pi b G y \text{ for non-self-energized gaskets} \quad (4.16.8)$$

$$W_{gs} = 0.0 \text{ for self-energized gaskets} \quad (4.16.9)$$

NOTE: Where significant axial force is required to compress the gasket during assembly of a joint containing a self-energizing gasket, the value of W_{gs} shall be taken as equal to that axial force. In addition, some self-energizing gaskets generate axial load due to their wedging action and this load shall be considered in setting the value of W_{gs} .

4.16.7 FLANGE DESIGN PROCEDURE

4.16.7.1 The procedure in this paragraph can be used to design circular integral, loose or reverse flanges, subject to internal or external pressure, and external loadings. The procedure incorporates both a strength check and a rigidity check for flange rotation.

4.16.7.2 The procedure to design a flange is shown below. (25)

Step 1. Determine the design pressure and temperature of the flange joint, and the external net-section axial force, F_A , and bending moment, M_E . If the pressure is negative, the absolute value of the pressure should be used in this procedure.

Step 2. Determine the design bolt loads for operating condition, W_o , and the gasket seating condition, W_g , and corresponding actual bolt area, A_b , from 4.16.6.

Step 3. Determine an initial flange geometry, in addition to the information required to determine the bolt load, the following geometric parameters are required:

- (a) The flange bore, B
- (b) The bolt circle diameter, C
- (c) The outside diameter of the flange, A
- (d) The flange thickness, t
- (e) The thickness of the hub at the large end, g_1
- (f) The thickness of the hub at the small end, g_0
- (g) The hub length, h

Step 4. Determine the flange stress factors using the equations in Tables 4.16.4 and 4.16.5.

Step 5. Determine the flange forces.

$$H_D = 0.785B^2P \quad (4.16.10)$$

$$H = 0.785G^2P \quad (4.16.11)$$

$$H_T = H - H_D \quad (4.16.12)$$

$$H_G = W_o - H \quad (4.16.13)$$

Step 6. Determine the flange moment for the operating condition using eq. (4.16.14) or eq. (4.16.15), as applicable. When specified by the user or the user's designated agent, the maximum bolt spacing (B_{smax}) and the bolt spacing correction factor (B_{sc}) shall be applied in calculating the flange moment for internal pressure using the equations in Table 4.16.11. The flange moment M_o for the operating condition and flange moment M_g for the gasket seating condition without correction for bolt spacing $B_{sc} = 1$ is used for the calculation of the rigidity index in Step 10. In these equations, h_D , h_T , and h_G are determined from Table 4.16.6. For integral and loose type flanges, the moment M_{oe} is calculated using eq. (4.16.16) where I and I_p in this equation are determined from Table 4.16.7. For reverse type flanges, the procedure to determine M_{oe} shall be agreed upon between the Designer and the Owner.

$$M_o = abs\left[\left((H_D h_D + H_T h_T + H_G h_G) B_{sc} + M_{oe}\right) F_{sr}\right] \text{ for internal pressure} \quad (4.16.14)$$

$$M_o = abs\left[\left(H_D(h_D - h_G) + H_T(h_T - h_G) + M_{oe}\right) F_{sr}\right] \text{ for external pressure} \quad (4.16.15)$$

$$M_{oe} = 4M_E \left[\frac{I}{0.3846I_p + I} \right] \left[\frac{h_D}{(C - 2h_D)} \right] + F_A h_D \quad (4.16.16)$$

Step 7. Determine the flange moment for the gasket seating condition using eq. (4.16.17) or eq. (4.16.18), as applicable.

$$M_g = \frac{W_g(C - G) B_{sc} F_{sr}}{2} \text{ for internal pressure} \quad (4.16.17)$$

$$M_g = W_g h_G F_{sr} \text{ for external pressure} \quad (4.16.18)$$

Step 8. Determine the flange stresses for the operating and gasket seating conditions using the equations in Table 4.16.8.

Step 9. Check the flange stress acceptance criteria. The two criteria shown below shall be evaluated. If the stress criteria are satisfied, go to Step 10. If the stress criteria are not satisfied, re-proportion the flange dimensions and go to Step 4.

(a) Allowable Normal Stress - The criteria to evaluate the normal stresses for the operating and gasket seating conditions are shown in Table 4.16.9.

(b) Allowable Shear Stresses - In the case of loose type flanges with lap, as shown in Figure 4.16.5 where the gasket is so located that the lap is subjected to shear, the shearing stress shall not exceed $0.8S_{no}$ or $0.8S_{ng}$, as applicable, for the material of the lap. In the case of welded flanges where the nozzle neck, vessel, or pipe wall extends near to the flange face and may form the gasket contact face, the shearing stress carried by the welds shall not exceed $0.8S_{no}$ or $0.8S_{ng}$, as applicable. The shearing stress shall be calculated for both the operating and gasket seating load cases. Similar situations where flange parts are subjected to shearing stresses shall be checked using the same requirement.

Step 10. Check the flange rigidity criterion in Table 4.16.10. If the flange rigidity criterion is satisfied, then the design is complete. If the flange rigidity criterion is not satisfied, then re-proportion the flange dimensions and go to Step 3. The flange moment M_o for the operating condition (see Step 6) and flange moment M_g for the gasket seating condition (see Step 7) without correction for bolt spacing $B_{sc} = 1$ is used for the calculation of the rigidity index.

(25) 4.16.8 SPLIT LOOSE TYPE FLANGES

Loose flanges split across a diameter and designed under the rules given in this paragraph may be used under the following provisions.

(a) When the flange consists of a single split flange or flange ring, it shall be designed as if it were a solid flange (without splits), using 200% of the total moment, $F_{sr} = 2.0$.

(b) When the flange consists of two split rings, each ring shall be designed as if it were a solid flange (without splits), using 75% of the total moment, $F_{sr} = 0.75$. The pair of rings shall be assembled so that the splits in one ring are 90 deg from the splits in the other ring.

(c) The flange split locations should preferably be midway between bolt holes.

(d) It is not a requirement that the flange rigidity rules of 4.6.10 be applied to split loose flanges.

4.16.9 NONCIRCULAR SHAPED FLANGES WITH A CIRCULAR BORE

The outside diameter, A , for a noncircular flange with a circular bore shall be taken as the diameter of the largest circle, concentric with the bore, inscribed entirely within the outside edges of the flange. The bolt loads, flange moments, and stresses shall be calculated in the same manner as for a circular flange using a bolt circle whose size is established by drawing a circle through the centers of the outermost bolts.

4.16.10 FLANGES WITH NUT STOPS

When flanges are designed per 4.16, or are fabricated to the dimensions of ASME B16.5 or other acceptable standards, except that the dimension $0.5(C - B) - g_1$ is decreased to provide a nut-stop, the fillet radius shall be as shown in Figures 4.16.3 and 4.16.4 except that:

(a) For flanges designed to this paragraph, the minimum thickness of the hub at the large end, g_1 , shall be the smaller of $2t_n$ or $4r_u$, but not less than 12 mm (0.5 in.).

(b) For ASME B16.5 or other standard flanges, the thickness of the hub at the small end, g_0 , shall be increased as necessary to provide a nut-stop.

4.16.11 JOINT ASSEMBLY PROCEDURES

(25)

Bolted joints should be assembled and bolted-up in accordance with a written procedure that has been demonstrated to be acceptable for similar joint configurations in similar services. Further guidance can be found in ASME PCC-1, Pressure Boundary Bolted Flange Joint Assembly.

4.16.12 EVALUATION OF EXTERNAL FORCES AND MOMENTS FOR FLANGED JOINTS WITH STANDARD FLANGES

External loads (forces and bending moments) may be evaluated for flanged joints with welding neck flanges chosen in accordance with 4.1.11.1(a), 4.1.11.1(g), and 4.1.11.3, using the following requirements:

(a) The vessel design pressure (corrected for the static head from liquid or bulk material acting on the flange) at the design temperature cannot exceed the pressure-temperature rating of the flange.

(b) The actual assembly bolt load (see 4.16.11) shall comply with ASME PCC-1, Nonmandatory Appendix O.

(c) The bolt material shall have an allowable stress equal to or greater than SA-193 B8 Cl. 2 at the specified bolt size and temperature.

(d) The combination of vessel design pressure, P (corrected for the static head from liquid or bulk material acting on the flange), with external moment and external axial force shall satisfy eq. (4.16.19). (The units of the variables in this equation shall be consistent with the pressure rating.)

$$16M_E + 4F_A G \leq \pi G^3 \left[(P_R - P) + F_M P_R \right] \quad (4.16.19)$$

4.16.13 NOMENCLATURE

(25)

A = outside diameter of the flange or, where slotted holes extend to the outside of the flange, the diameter to the bottom of the slots

A_b = total cross-sectional area of all the bolts based on the smaller of
(a) the root diameter of the thread
(b) the least diameter of any unthreaded portion

ASME PCC-1, Appendix H contains root areas for common bolt sizes that may be used.

A_m = total minimum required cross-sectional area of the bolts

a = nominal bolt diameter

B = inside diameter of the flange. When $B < 20g_1$, B_1 may be used for B in the equation for the longitudinal stress
 $B_1 = B + g_1$ for loose type flanges and for integral type flanges that have a value of f less than 1.0, (although a minimum value of $f = 1.0$ is permitted). B_1 is equal to $B + g_0$ for integral type flanges when $f \geq 1.0$.

B^* = inside diameter of the reverse flange

B_s = bolt spacing. The bolt spacing may be taken as the bolt circle circumference divided by the number of bolts or as the chord length between adjacent bolt locations.

$B_{s\max}$ = maximum bolt spacing

B_{sc} = bolt spacing correction factor

b = effective gasket contact width

b_0 = basic gasket seating width

- C = bolt circle diameter
 C_{ul} = conversion factor for length, $C_{ul} = 1.0$ for U.S. Customary Units and $C_{ul} = 25.4$ for Metric Units
 d = flange stress factor
 d_r = flange stress factor d for a reverse type flange
 E_{yg} = modulus of elasticity at the gasket seating load case temperature
 E_{yo} = modulus of elasticity at the operating load case temperature
 e = flange stress factor
 e_r = flange stress factor e for a reverse type flange
 F = flange stress factor for integral type flanges
 F_A = value of the external tensile net-section axial force. Compressive net-section forces are to be neglected and for that case, F_A should be taken as equal to zero.
 F_L = flange stress factor for loose type flanges
 F_M = moment factor, in accordance with Table 4.16.12
 F_s = flange stress factor for welded slip-on flanges
 F_{sr} = moment factor used to design split rings (see 4.16.8), $F_{sr} = 1.0$ for non-split rings
 f = hub stress correction factor for integral flanges
 G = diameter at the location of the gasket load reaction (see Figure 4.16.8)
 G_{avg} = average of the hub thicknesses g_1 and g_0
 G_c = outside diameter of the gasket contact area (see Figure 4.16.8)
 g_0 = thickness of the hub at the small end
 (a) for integral type flanges per Figure 4.16.1(a), $g_0 = t_n$
 (b) for other integral type flanges, g_0 = the minimum of t_n or the thickness of the hub at the small end
 g_1 = thickness of the hub at the large end
 g_2 = thickness of the pipe in a welded slip-on flange, $g_2 = t_n$
 H = total hydrostatic end force
 H_D = total hydrostatic end force on the area inside of the flange
 H_G = gasket load for the operating condition
 H_T = difference between the total hydrostatic end force and hydrostatic end force on the area inside the flange
 h = hub length
 h_n = the axial length of the cylinder to which the flange ring is attached. This is measured from the back face of flange to the other end of the cylinder. This is used when $g_0 = g_1$.
 h_o = hub length parameter
 h_{or} = hub length parameter for a reverse flange
 h_p = effective hub length used to determine I_p
 h_D = moment arm for load H_D
 h_G = moment arm for load H_G
 h_T = moment arm for load H_T
 I = bending moment of inertia of the flange cross-section
 I_p = polar moment of inertia of the flange cross-section
 J = flange rigidity index
 K = ratio of the flange outside diameter to the flange inside diameter
 K_R = rigidity index factor
 L = flange stress factor
 L_r = flange stress factor L for a reverse type flange
 M_E = absolute value of the external net-section bending moment
 M_g = flange design moment for the gasket seating condition
 M_o = flange design moment for the operating condition
 M_{oe} = component of the flange design moment resulting from a net section bending moment and/or axial force
 m = factor for the gasket operating condition
 N = gasket contact width, $N = 0.0$ for self-energizing gaskets
 P = design pressure
 P_R = flange pressure rating at design temperature
 r_1 = radius to be at least $0.25g_1$ but not less than 5 mm (0.1875 in.)
 r_u = radius of the undercut on a flange with nut stops
 S_{bg} = allowable stress from Annex 3-A for the bolt evaluated at the gasket seating temperature
 S_{bo} = allowable stress from Annex 3-A for the bolt evaluated at the design temperature
 S_{fg} = allowable stress from Annex 3-A for the flange evaluated at the gasket seating temperature

- S_{fo} = allowable stress from Annex 3-A for the flange evaluated at the design temperature
 S_{ng} = allowable stress from Annex 3-A for the nozzle neck, vessel, or pipe evaluated at the gasket seating temperature
 S_{no} = allowable stress from Annex 3-A for the nozzle neck, vessel, or pipe evaluated at the design temperature
 S_H = flange hub stress
 S_R = flange radial stress
 S_T = flange tangential stress
 S_{T1} = flange tangential stress at the outside diameter of a reverse flange
 S_{T2} = flange tangential stress at the inside diameter of a reverse flange
 T = flange stress factor
 T_r = flange stress factor T for a reverse flange
 t = flange thickness, including the facing thickness or the groove depth if either do not exceed 2 mm (0.0625 in.); otherwise, the facing thickness or groove depth is not included in the overall flange thickness
 t_g = uncompressed gasket thickness
 t_n = nominal thickness of the shell, pipe, or nozzle to which the flange is attached
 t_x = is $2g_o$ when the design is calculated as an integral flange, or two times the minimum required thickness of the shell or nozzle wall when the design is based on a loose flange, but not less than 6 mm (0.25 in.)
 U = flange stress factor
 U_r = flange stress factor U for a reverse type flange
 V = flange stress factor for integral type flanges
 V_L = flange stress factor for loose type flanges
 V_s = flange stress factor for welded slip-on flanges
 W_g = design bolt load for the gasket seating condition
 W_o = design bolt load for the operating condition
 w = width of the raised face when narrower than the gasket width
 y = factor for the gasket seating condition
 Y = flange stress factor
 Y_r = flange stress factor Y for a reverse type flange
 Z = flange stress factor

4.16.14 TABLES

Table 4.16.1
Gasket Factors for Determining the Bolt Loads

(25)

Gasket Material	Gasket Factor, m	Min. Design Seating Stress, y , MPa (psi)	Facing Sketch in Table 4.16.3
Self-energizing types (O rings, metallic, elastomer, other gasket types considered as self-sealing)	0	0 in. 0	...
Elastomers without fabric or high percent of mineral fiber:			
below 75 A Shore Durometer	0.50	0	(1a), (1b), (1c),
75 A or higher Shore Durometer	1.00	0 1.4 (200)	(1d)
Mineral fiber with suitable binder for operating conditions:			
3.2 mm ($\frac{1}{8}$ in.) thick	2.00	11 (1,600)	(1a), (1b), (1c), (1d)
1.6 mm ($\frac{1}{16}$ in.) thick	2.75	26 (3,700)	
0.8 mm ($\frac{1}{32}$ in.) thick	3.50	45 (6,500)	
Elastomers with cotton fabric insertion	1.25	2.8 (400)	(1a), (1b), (1c), (1d)
Elastomers with mineral fiber insertion (with or without wire reinforcement):			
3-ply	2.25	15 (2,200)	(1a), (1b), (1c), (1d)
2-ply	2.50	20 (2,900)	
1-ply	2.75	26 (3,700)	
Vegetable fiber	1.75	7.6 (1,100)	(1a), (1b), (1c), (1d)

Table 4.16.1
Gasket Factors for Determining the Bolt Loads (Cont'd)

Gasket Material	Gasket Factor, <i>m</i>	Min. Design Seating Stress, <i>y</i> , MPa (psi)	Facing Sketch in Table 4.16.3
Spiral-wound metal, mineral fiber filler:			
Carbon steel	2.50	69 (10,000)	
Stainless steel, Monel, and nickel-base alloy	3.00	69 (10,000)	(1a), (1b)
Corrugated metal, mineral fiber inserted, or corrugated metal, jacketed mineral fiber filled:			
Soft aluminum	2.50	20 (2,900)	
Soft copper or brass	2.75	26 (3,700)	
Iron or soft steel	3.00	31 (4,500)	
Monel or 4%–6% chrome	3.25	38 (5,500)	
Stainless steels and nickel-base alloys	3.50	45 (6,500)	(1a), (1b)
Corrugated metal:			
Soft aluminum	2.75	26 (3,700)	
Soft copper or brass	3.00	31 (4,500)	
Iron or soft steel	3.25	38 (5,500)	
Monel or 4%–6% chrome	3.50	45 (6,500)	(1a), (1b), (1c),
Stainless steels and nickel-base alloys	3.75	52 (7,600)	(1d)
Flat metal, jacketed mineral fiber filled:			
Soft aluminum	3.25	38 (5,500)	
Soft copper or brass	3.50	45 (6,500)	
Iron or soft steel	3.75	52 (7,600)	
Monel	3.50	55 (8,000)	
4%–6% chrome	3.75	62 (9,000)	(1a), (1b), (1c),
Stainless steels and nickel-base alloys	3.75	62 (9,000)	(1d)
Grooved metal:			
Soft aluminum	3.25	38 (5,500)	
Soft copper or brass	3.50	45 (6,500)	
Iron or soft steel	3.75	52 (7,600)	
Monel or 4%–6% chrome	3.75	62 (9,000)	(1a), (1b), (1c),
Stainless steels and nickel-base alloys	4.25	70 (10,100)	(1d)
Sold flat metal:			
Soft aluminum	4.00	61 (8,800)	
Soft copper or brass	4.75	90 (13,000)	
Iron or soft steel	5.50	124 (18,000)	
Monel or 4%–6% chrome	6.00	150 (21,800)	(1a), (1b), (1c),
Stainless steels and nickel-base alloys	6.50	180 (26,000)	(1d)
Ring joint:			
Iron or soft steel	5.50	124 (18,000)	
Monel or 4%–6% chrome	6.00	150 (21,800)	
Stainless steel and nickel-base alloys	6.50	180 (26,000)	(2)

GENERAL NOTE: This table gives a list of commonly used gasket materials and contact facings with suggested values of *m* and *y* that have generally proved satisfactory in actual service when using effective gasket seating width *b*. The design values and other details given in this table are suggested only and are not mandatory.

Table 4.16.2
Recommended Minimum Gasket Contact Width

Gasket Type	Gasket Contact Width, N				
	Gasket Outside Diameter				
	<150 mm (6 in.)	<300 mm (12 in.)	<600 mm (24 in.)	<900 mm (36 in.)	900 mm (36 in.) and Over
Sheet gaskets including laminated sheets gaskets with or without a metal core	9 mm (0.375 in.)	12 mm (0.5 in.)	16 mm (0.625 in.)	16 mm (0.625 in.)	19 mm (0.75 in.)
Preformed composite gaskets including spiral wound, jacketed, and solid flat metal gaskets	6 mm (0.25 in.)	9 mm (0.375 in.)	12 mm (0.5 in.)	16 mm (0.625 in.)	16 mm (0.625 in.)

Table 4.16.3
Effective Gasket Width for Determining the Bolt Loads

Facing Sketch	Facing Sketch Detail (Exaggerated)	Basic Gasket Seating Width, b_o
1a		$\frac{N}{2}$
1b		
1c		$\min \left[\frac{w + t_g}{2}, \frac{w + N}{4} \right]$
1d		
2		$\frac{w}{8}$

GENERAL NOTE: The gasket factors listed in this table apply only to flanged joints in which the gasket is contained entirely within the inner edges of the bolt holes.

NOTE:

- (1) Raised face on one side or the combination of raised faces on both sides is sufficiently high that the gasket will not be compressed outside of the raised face region. If the gasket is compressed, then facing 1(a) or 1(b) should be used, as appropriate.

(25)

(25)

Table 4.16.4
Flange Stress Factors Equations Involving Diameter

Flange Type	Stress Factors Involving Diameter
Integral-type flange, welded slip-on-type flange, and loose-type flange with a hub	$K = \frac{A}{B}$
	$Y = \frac{1}{K-1} \left[0.66845 + 5.71690 \left(\frac{K^2 \log_{10} K}{K^2 - 1} \right) \right]$
	$T = \frac{K^2 (1 + 8.55246 \log_{10} K) - 1}{(1.04720 + 1.9448 K^2)(K - 1)}$
	$U = \frac{K^2 (1 + 8.55246 \log_{10} K) - 1}{1.36136 (K^2 - 1)(K - 1)}$
	$Z = \frac{(K^2 + 1)}{(K^2 - 1)}$
	$L = \frac{te + 1}{T} + \frac{t^3}{d}$
	$e = \frac{F}{h_o} \text{ for integral-type flanges}$
	$e = \frac{F_1}{h_o} \text{ for loose-type flanges with a hub}$
	$e = \frac{F_s}{h_0} \text{ for welded slip-on flanges}$
	$d = \frac{U g_0^2 h_o}{V} \text{ for integral-type flanges}$
	$d = \frac{U g_0^2 h_o}{V_L} \text{ for loose-type flanges with a hub}$
	$d = \frac{U g_0^2 h_0}{V_s} \text{ for welded slip-on flanges}$
	$h_o = \sqrt{B g_0}$
	$X_g = \frac{g_1}{g_0}$
	$X_{g2} = \frac{g_2}{g_0}$
	$X_h = 2.0 \text{ for integral type flanges with } g_1 = g_0$
	$X_h = \frac{h}{h_o} \text{ for all others}$

Table 4.16.4
Flange Stress Factors Equations Involving Diameter (Cont'd)

Flange Type	Stress Factors Involving Diameter
Reverse integral-type flange and reverse loose-type flanges with a hub	The parameters K , T , U , Y , and Z are determined using the equations for integral- and loose-type flanges with:
	$K = \frac{A}{B^*}$
	Then, the reverse flange parameters are computed as follows:
	$Y_r = \alpha_r Y$
	$T_r = \left(\frac{Z + 0.3}{Z - 0.3} \right) \alpha_r T$
	$U_r = \alpha_r U$
	$L_r = \frac{(te_r + 1)}{T_r} + \frac{t^3}{d_r}$
	$\alpha_r = \frac{1}{K^2} \left(1 + \frac{0.668(K + 1)}{Y} \right)^Y$
	$e_r = \frac{F}{h_{or}} \text{ for integral-type flanges}$
	$e_r = \frac{F_L}{h_{or}} \text{ for loose-type flanges with a hub}$
	$d_r = \frac{U_r g_0^2 h_{or}}{V} \text{ for integral-type flanges}$
	$d_r = \frac{U_r g_0^2 h_{or}}{V_L} \text{ for loose-type flanges with a hub}$
	$h_{or} = \sqrt{A g_0}$
	$X_g = \frac{g_1}{g_0}$
	$X_h = 2.0 \text{ for integral type flanges with } g_1 = g_0$
	$X_h = \frac{h}{h_{or}} \text{ for all others}$

(25)

Table 4.16.5
Flange Stress Factor Equations

Flange Type	Stress Factors
Integral-type flange, reverse integral-type flange	$F = \left[\begin{aligned} &0.897697 - 0.297012 \ln X_g + 9.5257(10^{-3}) \ln X_h + \\ &0.123586(\ln X_g)^2 + 0.0358580(\ln X_h)^2 - 0.194422(\ln X_g)(\ln X_h) - \\ &0.0181259(\ln X_g)^3 + 0.0129360(\ln X_h)^3 - \\ &0.0377693(\ln X_g)(\ln X_h)^2 + 0.0273791(\ln X_g)^2(\ln X_h) \end{aligned} \right]$ For $0.1 \leq X_h \leq 0.5$ $V = \left[\begin{aligned} &0.500244 + \frac{0.227914}{X_g} - 1.87071X_h - \frac{0.344410}{X_g^2} + 2.49189X_h^2 + \\ &0.873446\left(\frac{X_h}{X_g}\right) + \frac{0.189953}{X_g^3} - 1.06082X_h^3 - 1.49970\left(\frac{X_h^2}{X_g}\right) + \\ &0.719413\left(\frac{X_h}{X_g^2}\right) \end{aligned} \right]$ For $0.5 < X_h \leq 2.0$ $V = \left(\begin{aligned} &0.0144868 - \frac{0.135977}{X_g} - \frac{0.0461919}{X_h} + \frac{0.560718}{X_g^2} + \frac{0.0529829}{X_h^2} + \\ &\frac{0.244313}{X_g X_h} + \frac{0.113929}{X_g^3} - \frac{0.00928265}{X_h^3} - \frac{0.0266293}{X_g X_h^2} - \frac{0.217008}{X_g^2 X_h} \end{aligned} \right)$ $f = \max \left[1.0, \frac{\left(\begin{aligned} &0.0927779 - 0.0336633X_g + 0.964176X_g^2 + \\ &0.0566286X_h + 0.347074X_h^2 - 4.18699X_h^3 \end{aligned} \right)}{\left(\begin{aligned} &1 - 5.96093(10^{-3})X_g + 1.62904X_h + \\ &3.49329X_h^2 + 1.39052X_h^3 \end{aligned} \right)} \right]$

Table 4.16.5
Flange Stress Factor Equations (Cont'd)

Flange Type	Stress Factors			
Welded slip-on flange	$F_S = \frac{\sum_{i,j} M_{i,j} X_{g2}^i X_h^j}{\sum_{i,j} N_{i,j} X_{g2}^i X_h^j}$			
	$V_S = \frac{\sum_{i,j} O_{i,j} X_{g2}^i X_h^j}{\sum_{i,j} P_{i,j} X_{g2}^i X_h^j}$			
	M =			
		<i>j</i> = 0	<i>j</i> = 1	<i>j</i> = 2
	<i>i</i> = 0	-0.33289917	1.89824507	0.51289982
	<i>i</i> = 1	5.33592441	-17.32691994	17.31106722
	<i>i</i> = 2	4.92214647	3.21485121	-1.76102532
	N =			
		<i>j</i> = 0	<i>j</i> = 1	<i>j</i> = 2
	<i>i</i> = 0	-0.0709392	0.30338397	1.74097066
	<i>i</i> = 1	0.54117473	-6.48801256	12.82177228
	<i>i</i> = 2	10.36447138	-6.91013557	2.9795664
	O =			
		<i>j</i> = 0	<i>j</i> = 1	<i>j</i> = 2
	<i>i</i> = 0	0.39926596	-1.09329423	0.79382644
	<i>i</i> = 1	-0.45359452	2.79772392	-2.02963598
	<i>i</i> = 2	0.26069646	-1.46342431	4.17498559
	P =			
		<i>j</i> = 0	<i>j</i> = 1	<i>j</i> = 2
	<i>i</i> = 0	0.000733115	-0.016647943	0.342479503
	<i>i</i> = 1	-0.02136683	0.173868541	-0.199696025
	<i>i</i> = 2	0.349898055	0.794317086	4.88408141

Table 4.16.5
Flange Stress Factor Equations (Cont'd)

Flange Type	Stress Factors
Loose-type flange with a hub, reverse loose-type flange with a hub	$F_L = \frac{\left(\begin{array}{l} 0.941074 + 0.176139(\ln X_g) - 0.188556(\ln X_h) + 0.0689847(\ln X_g)^2 + \\ 0.523798(\ln X_h)^2 - 0.513894(\ln X_g)(\ln X_h) \end{array} \right)}{\left(\begin{array}{l} 1 + 0.379392(\ln X_g) + 0.184520(\ln X_h) - 0.00605208(\ln X_g)^2 - \\ 0.00358934(\ln X_h)^2 + 0.110179(\ln X_g)(\ln X_h) \end{array} \right)}$ For $0.1 \leq X_h \leq 0.25$ $\ln[V_L] = \left[\begin{array}{l} 6.57683 - 0.115516X_g + 1.39499\sqrt{X_g}(\ln X_g) + \\ 0.307340(\ln X_g)^2 - 8.30849\sqrt{X_g} + 2.62307(\ln X_g) + \\ 0.239498X_h(\ln X_h) - 2.96125(\ln X_h) + \frac{7.035052(10^{-4})}{X_h} \end{array} \right]$ For $0.25 < X_h \leq 0.50$ $V_L = \left[\begin{array}{l} 1.56323 - 1.80696(\ln X_g) - \frac{1.33458}{X_h} + 0.276415(\ln X_g)^2 + \\ \frac{0.417135}{X_h^2} + \frac{1.39511(\ln X_g)}{X_h} + 0.0137129(\ln X_g)^3 + \\ \frac{0.0943597}{X_h^3} - \frac{0.402096(\ln X_g)}{X_h^2} - \frac{0.101619(\ln X_g)^2}{X_h} \end{array} \right]$ For $0.50 < X_h \leq 1.0$ $V_L = \left(\begin{array}{l} -0.0213643 - \frac{0.0763597}{X_g} + \frac{0.1029900}{X_h} + \frac{0.725776}{X_g^2} - \frac{0.160603}{X_h^2} - \\ \frac{0.0918061}{X_g \cdot X_h} + \frac{0.472277}{X_g^3} + \frac{0.0873530}{X_h^3} + \frac{0.527487}{X_g \cdot X_h^2} - \frac{0.980209}{X_g^2 \cdot X_h} \end{array} \right)$ For $1.0 < X_h \leq 2.0$ $V_L = \left[\begin{array}{l} 7.96687(10^{-3}) - \frac{0.220518}{X_g} + \frac{0.0602652}{X_h} + \frac{0.619818}{X_g^2} - \frac{0.223212}{X_h^2} + \\ \frac{0.421920}{X_g \cdot X_h} + \frac{0.0950195}{X_g^3} + \frac{0.209813}{X_h^3} - \frac{0.158821}{X_g \cdot X_h^2} - \frac{0.242056}{X_g^2 \cdot X_h} \end{array} \right]$ $f = 1.0$

Table 4.16.6
Moment Arms for Flange Loads for the Operating Condition

(25)

Flange Type	h_D	h_T	h_G
Integral and welded slip-on-type flanges	$\frac{C - B - g_1}{2}$	$\frac{1}{2} \left(\frac{C - B}{2} + h_G \right)$	$\frac{C - G}{2}$
Loose-type lap joint flanges	$\frac{C - B}{2}$	$\frac{C - G}{2}$	$\frac{C - G}{2}$
Reverse integral-type flanges	$\frac{C + g_1 - 2g_0 - B}{2}$	$\frac{1}{2} \left(C - \frac{B + G}{2} \right)$	$\frac{C - G}{2}$
Reverse loose-type flanges	$\frac{C - B}{2}$	$\frac{1}{2} \left(C - \frac{B + G}{2} \right)$	$\frac{C - G}{2}$

Table 4.16.7
Flange Moments of Inertia

(25)

Flange Type	I	I_p
Integral-type flange	$I = \frac{0.0874 L g_0^2 h_0 B}{V}$	$I_p = \max \left\{ K_{AB} + K_{CD}, A_R t^3 \left[\frac{1}{3} - 0.21 \left(\frac{t}{A_R} \right) \left(1 - \frac{1}{12} \left(\frac{t}{A_R} \right)^4 \right) \right] \right\}$ $K_{AB} = \left(A_A B_B^3 \right) \left[\frac{1}{3} - 0.21 \left(\frac{B_B}{A_A} \right) \left(1 - \frac{1}{12} \left(\frac{B_B}{A_A} \right)^4 \right) \right]$ $K_{CD} = \left(C_C D_{DG}^3 \right) \left[\frac{1}{3} - 0.105 \left(\frac{D_{DG}}{C_C} \right) \left(1 - \frac{1}{192} \left(\frac{D_{DG}}{C_C} \right)^4 \right) \right]$ $A_R = 0.5(A - B)$ $G_{avg} = 0.5(g_0 + g_1)$ For integral-type flanges with $g_1 = g_0$ $h_p = \max \left\{ 0.35 g_0, \min \left[3 g_0, h_n, 0.78 (0.5 B g_0)^{0.5} \right] \right\}$ For all others $h_p = h$ If $t \geq G_{avg}$ $A_A = A_R, B_B = t, C_C = h_p, D_{DG} = G_{avg}$ If $t < G_{avg}$ $A_A = h_p + t, B_B = G_{avg}, C_C = A_R - G_{avg}, D_{DG} = t$
Loose-type flange with a hub	$I = \frac{0.0874 L g_0^2 h_0 B}{V_L}$	
Welded slip-on-type flange	$I = \frac{0.0874 L g_0^2 h_0 B}{V_S}$	

Table 4.16.7
Flange Moments of Inertia (Cont'd)

Flange Type	I	I_p
Loose-type flange without a hub	$I = \frac{Bt^3 \ln K}{24}$	$I_p = A_R t^3 \left\{ \frac{1}{3} - 0.21 \left(\frac{t}{A_R} \right) \left[1 - \frac{1}{12} \left(\frac{t}{A_R} \right)^4 \right] \right\}$ $A_R = 0.5[A - B]$

(25)

Table 4.16.8
Flange Stress Equations

Flange Type	Stress Equations	
	Operating Condition	Gasket Seating Conditions
Integral-type flange, welded slip-on-type flange, or loose-type flange with a hub	$S_H = \frac{f M_o}{L_r g_1^2 B}$ $S_R = \frac{(1.33te + 1) M_o}{L t^2 B}$ $S_T = \frac{Y M_o}{t^2 B} - Z S_R$	$S_H = \frac{f M_g}{L_r g_1^2 B}$ $S_R = \frac{(1.33te + 1) M_g}{L t^2 B}$ $S_T = \frac{Y M_g}{t^2 B} - Z S_R$
Loose-type flange without a hub	$S_T = \frac{Y M_o}{t^2 B}$	$S_T = \frac{Y M_g}{t^2 B}$
Reverse integral-type flange or reverse loose-type flange with a hub	$S_H = \frac{f M_o}{L_r g_1^2 B^*}$ $S_R = \frac{(1.33te_r + 1) M_o}{L_r t^2 B^*}$ $S_{T1} = \frac{Y_r M_o}{t^2 B^*} - \frac{Z S_R (0.67te_r + 1)}{(1.33te_r + 1)}$ $S_{T2} = \left(Y - \frac{2K^2 (0.67te_r + 1)}{(K^2 - 1) L_r} \right) \frac{M_o}{t^2 B^*}$	$S_H = \frac{f M_g}{L_r g_1^2 B^*}$ $S_R = \frac{(1.33te_r + 1) M_g}{L_r t^2 B^*}$ $S_{T1} = \frac{Y_r M_g}{t^2 B^*} - \frac{Z S_R (0.67te_r + 1)}{(1.33te_r + 1)}$ $S_{T2} = \left(Y - \frac{2K^2 (0.67te_r + 1)}{(K^2 - 1) L_r} \right) \frac{M_g}{t^2 B^*}$
Reverse loose-type flange without a hub	$S_T = \frac{Y M_o}{t^2 B^*}$	$S_T = \frac{Y M_g}{t^2 B^*}$

(25)

Table 4.16.9
Flange Stress Acceptance Criteria

Flange Type	Stress Acceptance Criteria	
	Operating Condition	Gasket Seating Conditions
Integral-type flange, welded slip-on-type flange, or loose-type flange with a hub	$S_H \leq \min[1.5S_{fo}, 2.5S_{no}] \text{ [Note (1)]}$ $S_H \leq 1.5S_{fo} \text{ [Note (2)]}$ $S_R \leq S_{fo}$ $S_T \leq S_{fo}$ $\frac{(S_H + S_R)}{2} \leq S_{fo}$ $\frac{(S_H + S_T)}{2} \leq S_{fo}$	$S_H \leq \min[1.5S_{fg}, 2.5S_{ng}] \text{ [Note (1)]}$ $S_H \leq 1.5S_{fg} \text{ [Note (2)]}$ $S_R \leq S_{fg}$ $S_T \leq S_{fg}$ $\frac{(S_H + S_R)}{2} \leq S_{fg}$ $\frac{(S_H + S_T)}{2} \leq S_{fg}$

Table 4.16.9
Flange Stress Acceptance Criteria (Cont'd)

Flange Type	Stress Acceptance Criteria	
	Operating Condition	Gasket Seating Conditions
Loose-type flange without a hub	$S_T \leq S_{fo}$	$S_T \leq S_{fg}$
Reverse integral-type flange or reverse loose-type flange with a hub	$S_H \leq 1.5S_{fo}$ $S_R \leq S_{fo}$ $S_{T1} \leq S_{fo}$ $\frac{(S_H + S_R)}{2} \leq S_{fo}$ $\frac{(S_H + S_{T1})}{2} \leq S_{fo}$ $S_{T2} \leq S_{fo}$	$S_H \leq 1.5S_{fg}$ $S_R \leq S_{fg}$ $S_{T1} \leq S_{fg}$ $\frac{(S_H + S_R)}{2} \leq S_{fg}$ $\frac{(S_H + S_{T1})}{2} \leq S_{fg}$ $S_{T2} \leq S_{fg}$
Reverse loose-type flanges	$S_T \leq S_{fo}$	$S_T \leq S_{fg}$

NOTES:

- (1) For integral flanges with hubs welded to a nozzle neck, pipe, or vessel shell.
 (2) For loose type flanges with a hub.
 (3) Flanges made of non-ductile material, such as cast iron, are not addressed by this section.

Table 4.16.10
Flange Rigidity Criterion

(25)

Flange Type	Rigidity Criterion	
	Operating Condition	Gasket Seating Conditions
Integral-type flange	$J = \frac{52.14VM_o}{LE_{yo}g_0^2K_Rh_o} \leq 1.0$	$J = \frac{52.14VM_g}{LE_{yg}g_0^2K_Rh_o} \leq 1.0$
Welded slip-on-type flange	$J = \frac{52.14V_S M_o}{LE_{yo}g_0^2K_Rh_o} \leq 1.0$	$J = \frac{52.14V_S M_g}{LE_{yg}g_0^2K_Rh_o} \leq 1.0$
Loose-type flange with a hub	$J = \frac{52.14V_L M_o}{LE_{yo}g_0^2K_Rh_o} \leq 1.0$	$J = \frac{52.14V_L M_g}{LE_{yg}g_0^2K_Rh_o} \leq 1.0$
Reverse integral-type flange	$J = \frac{52.14VM_o}{L_rE_{yo}g_0^2K_Rh_o} \leq 1.0$	$J = \frac{52.14VM_g}{L_rE_{yg}g_0^2K_Rh_o} \leq 1.0$
Reverse loose-type flange with a hub	$J = \frac{52.14V_L M_o}{L_rE_{yo}g_0^2K_Rh_o} \leq 1.0$	$J = \frac{52.14V_L M_g}{L_rE_{yg}g_0^2K_Rh_o} \leq 1.0$
Loose-type and reverse loose-type flange without a hub	$J = \frac{109.4M_o}{E_{yo}t^3K_R(\ln K)} \leq 1.0$	$J = \frac{109.4M_g}{E_{yg}t^3K_R(\ln K)} \leq 1.0$

GENERAL NOTES:

- (a) For an integral type flange, $K_R = 0.3$ unless other values are specified by the user.
 (b) For a loose type flange with or without a hub, $K_R = 0.2$ unless other values are specified by the user.

Table 4.16.11
Bolt Spacing Equations

Flange Type	Bolt Spacing Factors
All	$B_{s\max} = 2a + \frac{6t}{m + 0.5}$ $B_{sc} = \max \left(1, \sqrt{\frac{B_s}{2a + t}} \right)$

Table 4.16.12
Moment Factor, F_M

Standard	Size Range	Flange Pressure Rating Class					
		150	300	600	900	1500	2500
ASME B16.5	≤NPS 12	1.2	0.5	0.5	0.5	0.5	0.5
	>NPS 12 and ≤NPS 24	1.2	0.5	0.5	0.3	0.3	...
ASME B16.47							
Series A	All	0.6	0.1	0.1	0.1	...	...
Series B	<NPS 48	[Note (1)]	[Note (1)]	0.13	0.13	...	...
	≥NPS 48	0.1	[Note (2)]	...	...	...	...

GENERAL NOTES:

- (a) The combinations of size ranges and flange pressure classes for which this Table gives no moment factor value are outside the scope of this table.
- (b) The designer should consider reducing the moment factor if the loading is primarily sustained in nature and the bolted flange joint operates at a temperature where gasket creep/relaxation will be significant.

NOTES:

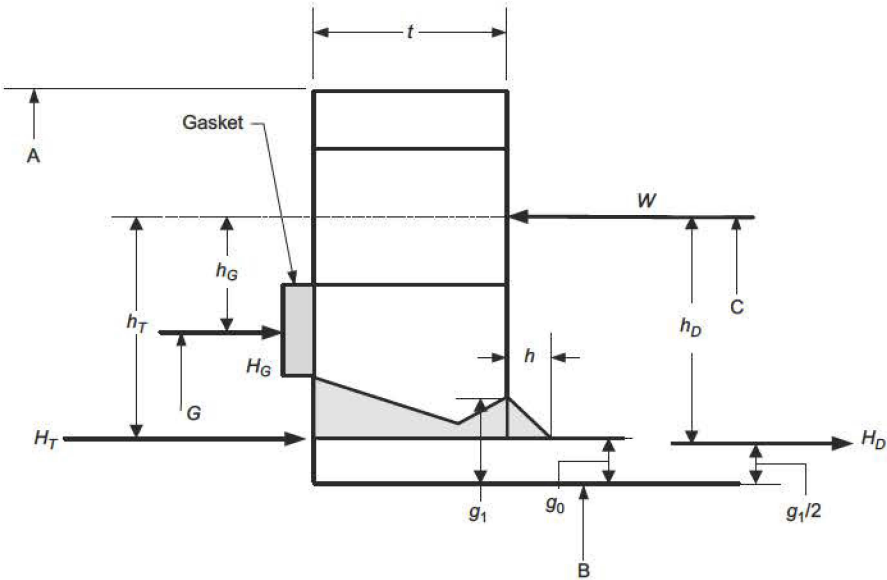
(1) $F_M = 0.1 + (48 - \text{NPS})/56$.

(2) $F_M = 0.1$ except NPS 60, Class 300, in which case $F_M = 0.03$.

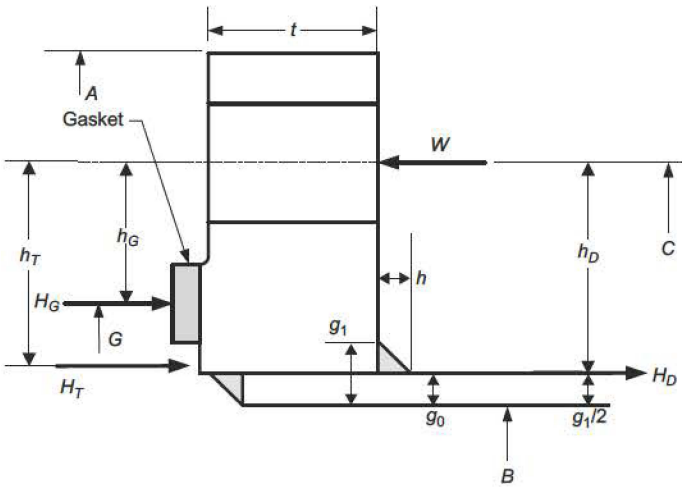
4.16.15 FIGURES

(25)

Figure 4.16.1
Integral Type Flanges

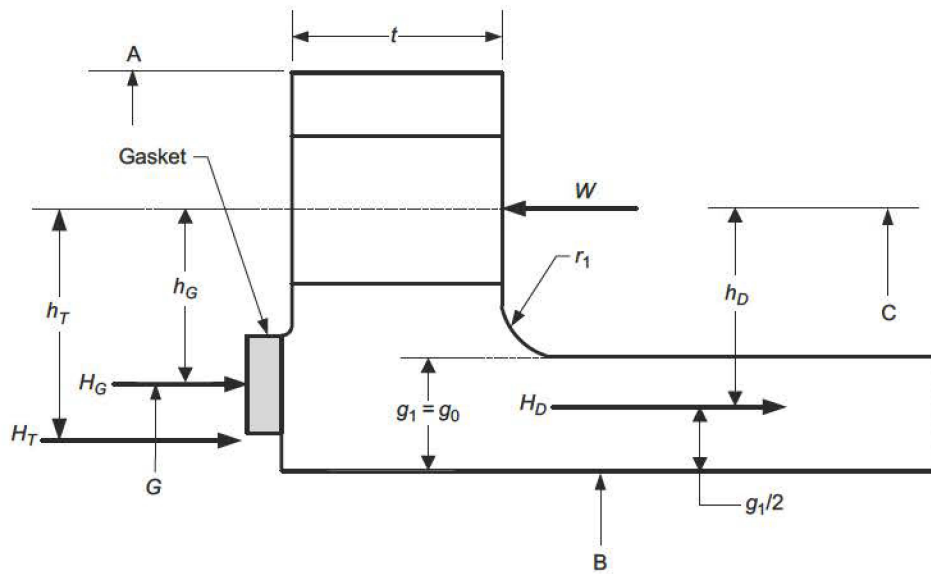


(a) Integral Flange With Full-Penetration Weld

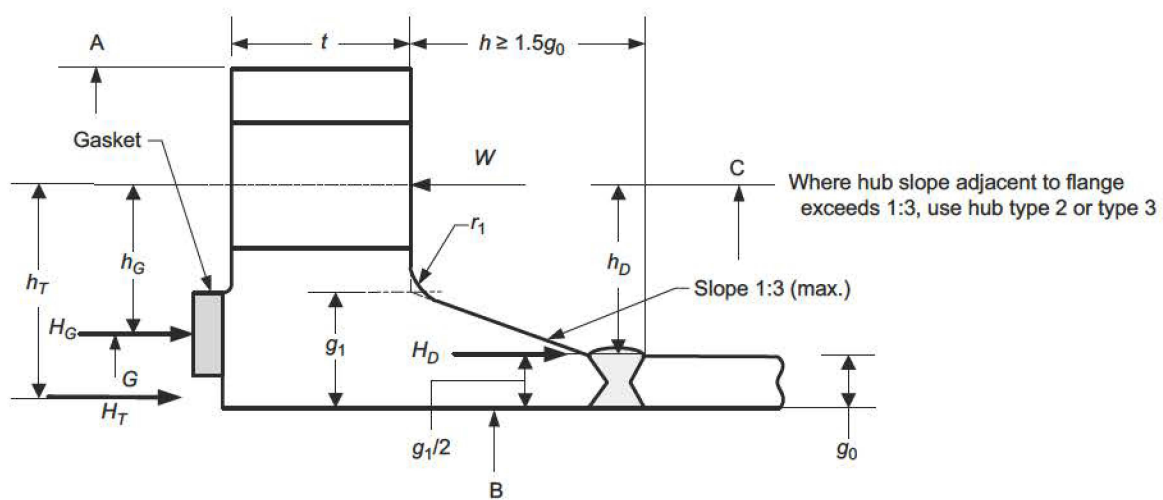


(b) Integral Flange With Partial-Penetration Weld

Figure 4.16.1
Integral Type Flanges (Cont'd)



(c) Integral Flange With $g_0 = g_1$



(d) Hub Type 1

Figure 4.16.1
Integral Type Flanges (Cont'd)

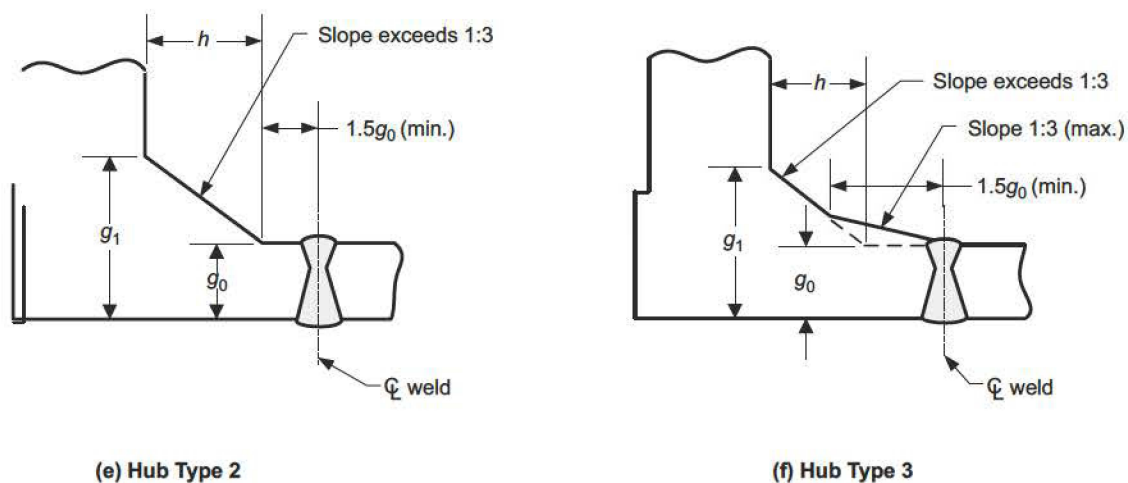


Figure 4.16.2
Welded Slip-on-Type Flanges

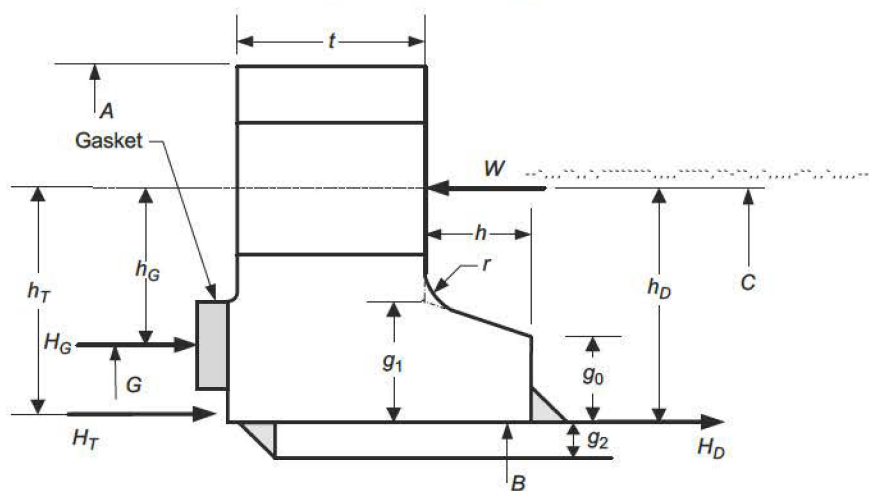
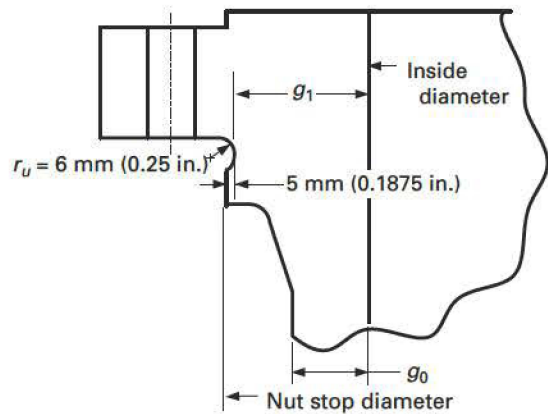
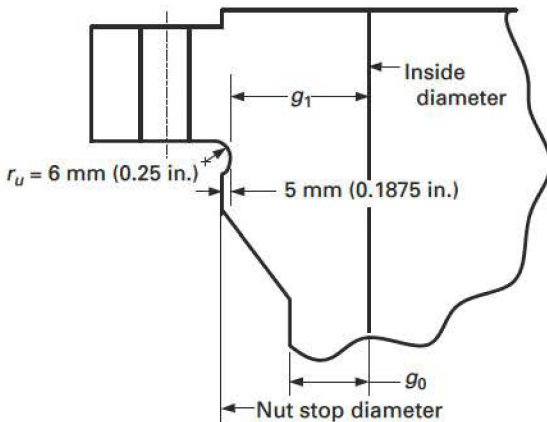


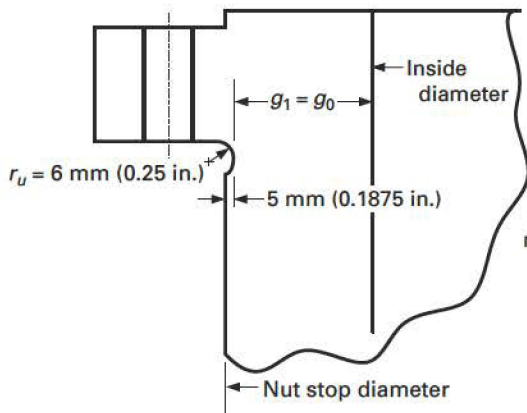
Figure 4.16.3
Integral Type Flanges With Nut Stops — Diameter Less Than or Equal to 450 mm (18 in.)



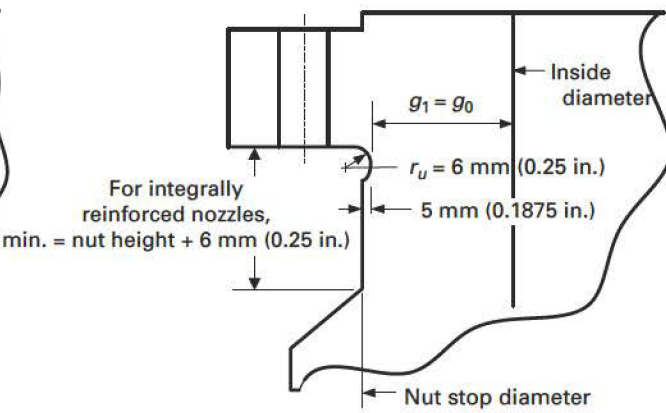
(a) Detail A



(b) Detail B

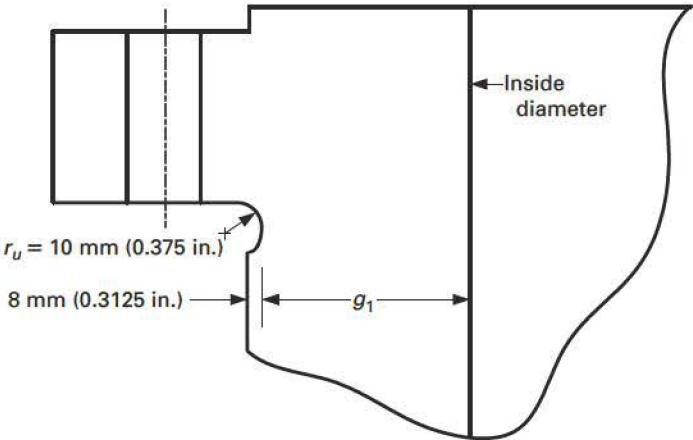


(c) Detail C



(d) Detail D

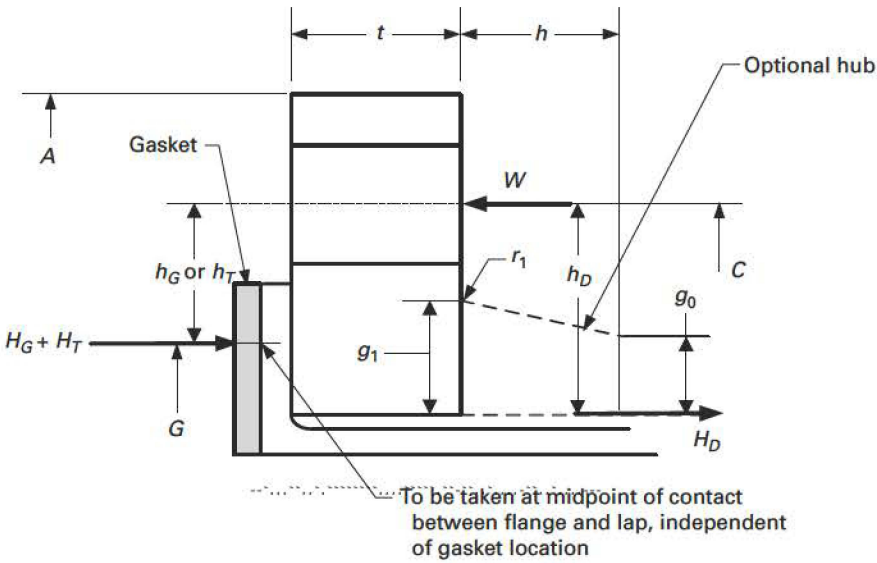
Figure 4.16.4
Integral Type Flanges With Nut Stops — Diameter Greater Than 450 mm (18 in.)



GENERAL NOTE: All other details per Figure 4.16.3.

Figure 4.16.5
Loose-Type Lap Joint Type Flanges

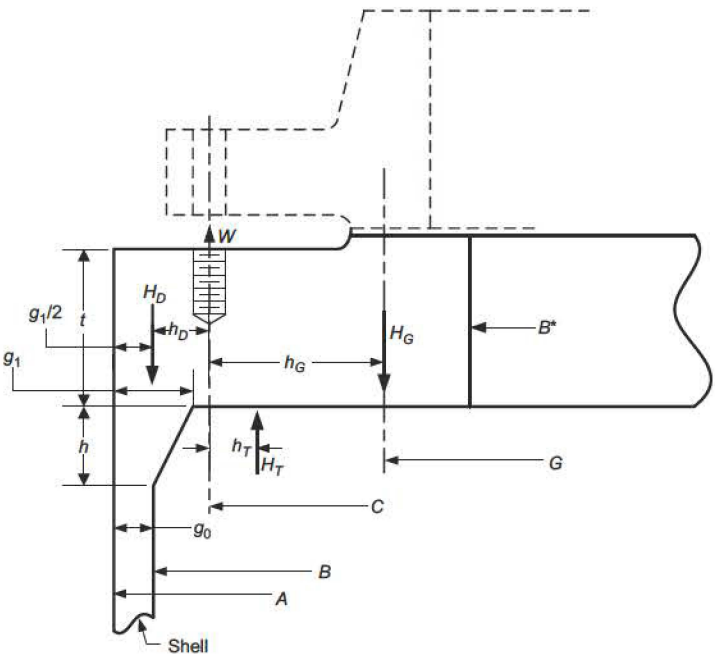
(25)



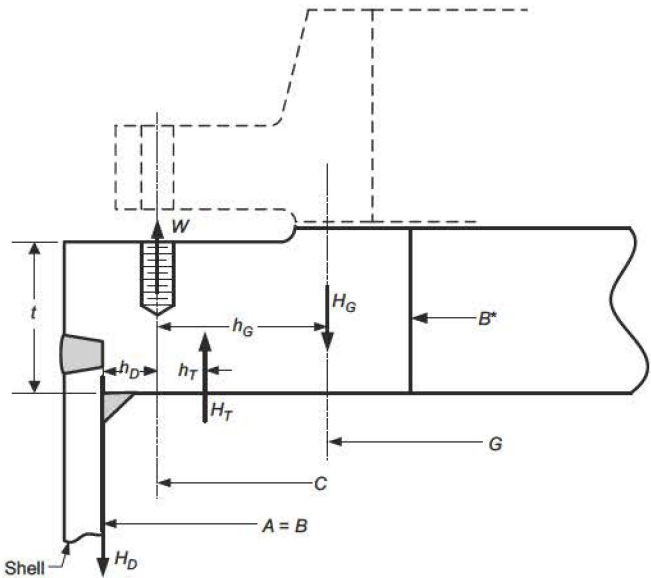
GENERAL NOTE: For hub tapers 6 deg or less, use $g_0 = g_1$ (loose-type flanges).

(25)

Figure 4.16.6
Integral-Type Reverse Flanges



(a) Reverse Flange With a Hub



(b) Reverse Flange Without a Hub

Figure 4.16.7
Loose-Type Reverse Flange

(25)

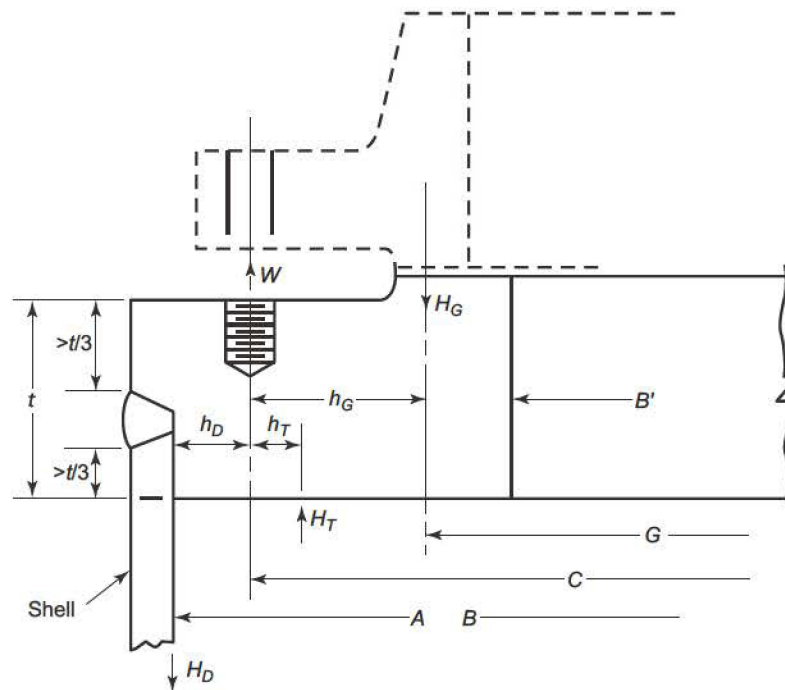
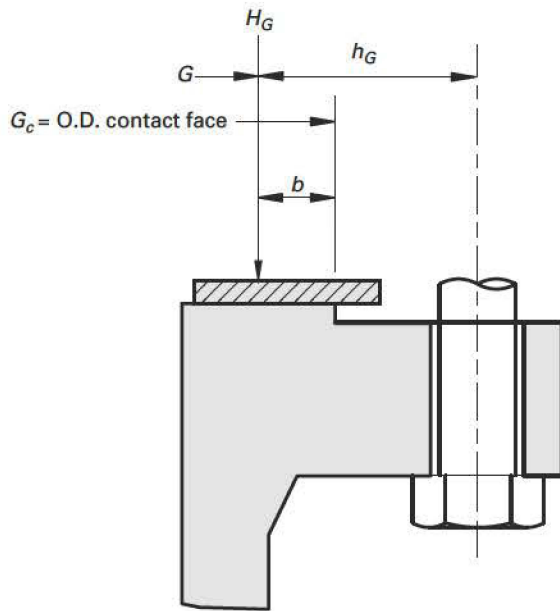
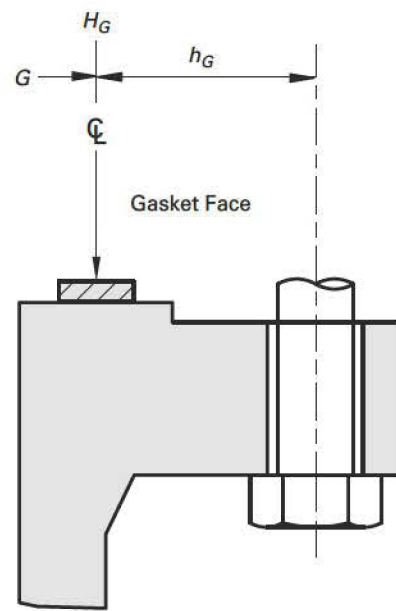


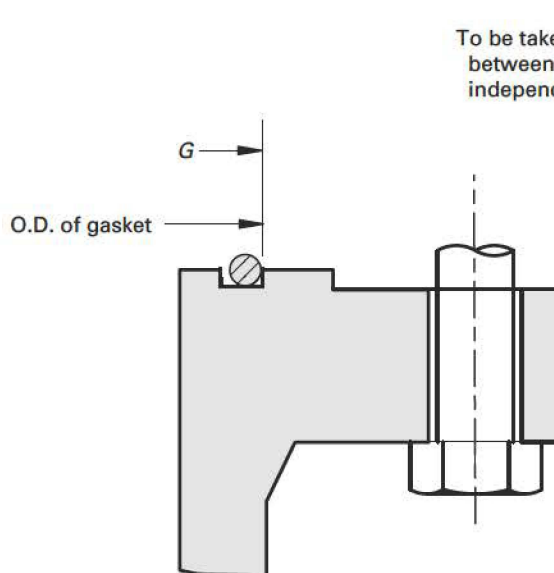
Figure 4.16.8
Location of Gasket Reaction Load Diameter



For $b_0 > 6 \text{ mm (0.25 in.)}$

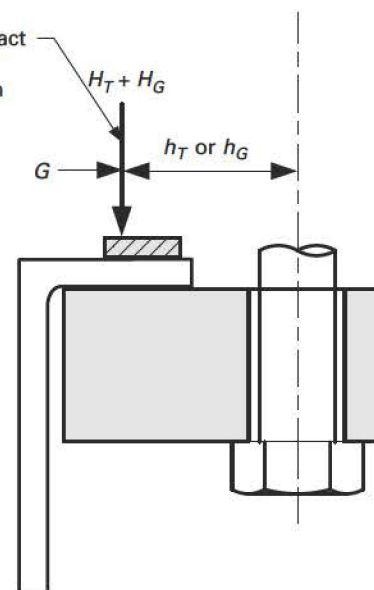


For $b_0 \leq 6 \text{ mm (0.25 in.)}$



Self-Energizing Gaskets

To be taken at midpoint of contact
between flange and lap,
independent of gasket location



Lap-Type Joints